

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Following are the functional requirements of the proposed Employee Laptop Request Management System.

Date	12 September 2026
Team ID	_____
Project Name	Employee Laptop Request Management System
Maximum Marks	4 Marks

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Laptop Request Submission	Provide a centralized Record Producer through the Service Portal for employees to submit laptop requests.
FR-2	Request Information Capture	Capture requester, employee email, request date, short description, manager, type of request and reason.
FR-3	Request Classification	Classify requests as Basic or Advance and capture Software Development, Personal Use or Other reason.
FR-4	Duplicate Request Prevention	Prevent the same employee from submitting more than one laptop request within the current calendar year.
FR-5	Manager Approval	Automatically send a submitted request to the referenced manager for approval or rejection.
FR-6	Status Management	Maintain Pending Approval, Approved and Rejected states according to the approval outcome.
FR-7	Email Notifications	Send approval-request notifications to managers and outcome notifications to employees.
FR-8	Role-Based Access	Allow IT employees to create/read/write requests and IT managers to view request records according to configured ACLs.
FR-9	Approval Relationship	Maintain the relationship between the Laptop Request Form and sysapproval_approver records for manager approval.
FR-10	Validation & Testing	Validate form behaviour, duplicate prevention, roles, ACLs, approval flow, notifications and update-set capture.

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a simple, consistent laptop request form so employees can submit requests with minimal confusion.
NFR-2	Security	Use ServiceNow roles and ACLs so access is restricted according to employee and manager responsibilities.
NFR-3	Reliability	Maintain consistent request, approval and status information throughout the request lifecycle.
NFR-4	Performance	Record Producer forms, request searches, approval actions and related records should respond efficiently under normal use.
NFR-5	Availability	Use ServiceNow platform capabilities to provide centralized access to the request application, subject to platform availability.
NFR-6	Scalability	Support growth in employees and laptop requests without changing the core request, approval and security design.

Project Design Phase-II

Technology Stack (Architecture & Stack)

Technical Architecture:

The solution is implemented as a ServiceNow-centered architecture. Employees submit requests through a Record Producer; the request is stored in the custom Laptop Request Form table, validated by backend logic, protected by roles and ACLs, and processed through Flow Designer for manager approval, status updates and email notifications.

Employee Laptop Request Management System - Technical Architecture

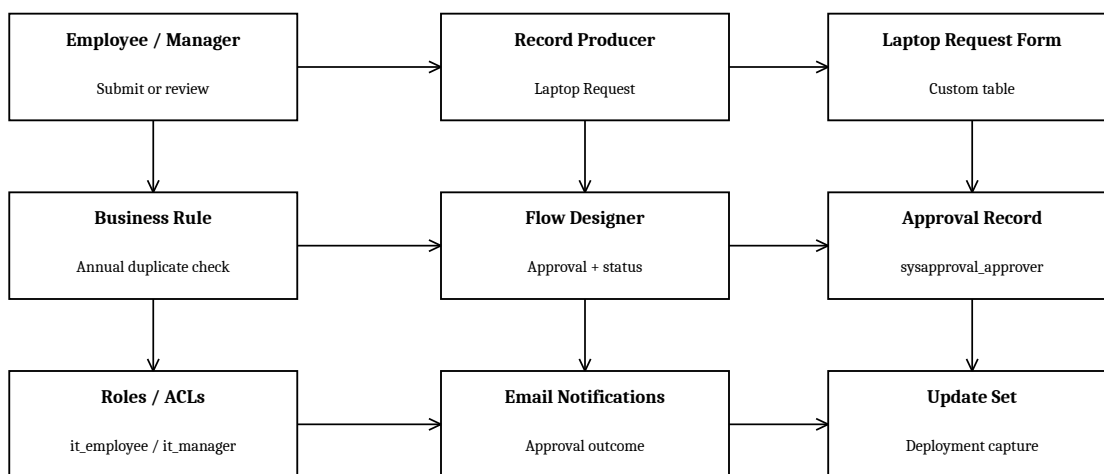


Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employees submit through the Service Portal Record Producer; managers review request records.	ServiceNow UI / Service Portal
2	Application Logic-1	Laptop requests are stored and managed in the custom scoped application table.	ServiceNow Custom Application
3	Application Logic-2	A before-insert Business Rule checks whether the requester already has a request in the current year.	ServiceNow Business Rule / GlideRecord
4	Application Logic-3	Flow Designer starts manager approval, handles outcome, updates status and sends email.	ServiceNow Flow Designer

Project Design Phase-II

Technology Stack (Architecture & Stack)

S.No	Component	Description	Technology
4	Security	Role-based permissions and table-level ACLs control create, read and write access.	ServiceNow Roles / ACLs
5	Approval	Manager approval records are maintained for the laptop request.	sysapproval_approver
6	Database	Stores requester, email, manager, type, reason, status and other request information.	ServiceNow Platform Database
7	File / Attachment Storage	ServiceNow platform storage can associate attachments with records if used.	ServiceNow Attachment Storage
8	External API	No project-specific external API is required for the implemented core process.	Not Applicable
9	Notifications	Approval requests and approval/rejection outcomes are communicated by email.	ServiceNow Email Notifications
10	Deployment	Application configuration is captured in an update set and moved through preview/commit/export.	ServiceNow Update Set

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Platform-Based	The project uses ServiceNow platform capabilities rather than a separate application framework.	ServiceNow platform
2	Security Implementations	Role-based access and ACLs restrict request operations according to employee and manager responsibility.	ServiceNow roles / ACLs
3	Scalable Architecture	The custom application separates request entry, backend validation, approval automation and managed data.	ServiceNow scoped application
4	Availability	The solution relies on the ServiceNow platform for hosted availability and centralized access.	ServiceNow cloud platform
5	Performance	Simple forms, controlled queries and automated processing support responsive operation under normal use.	ServiceNow platform capabilities

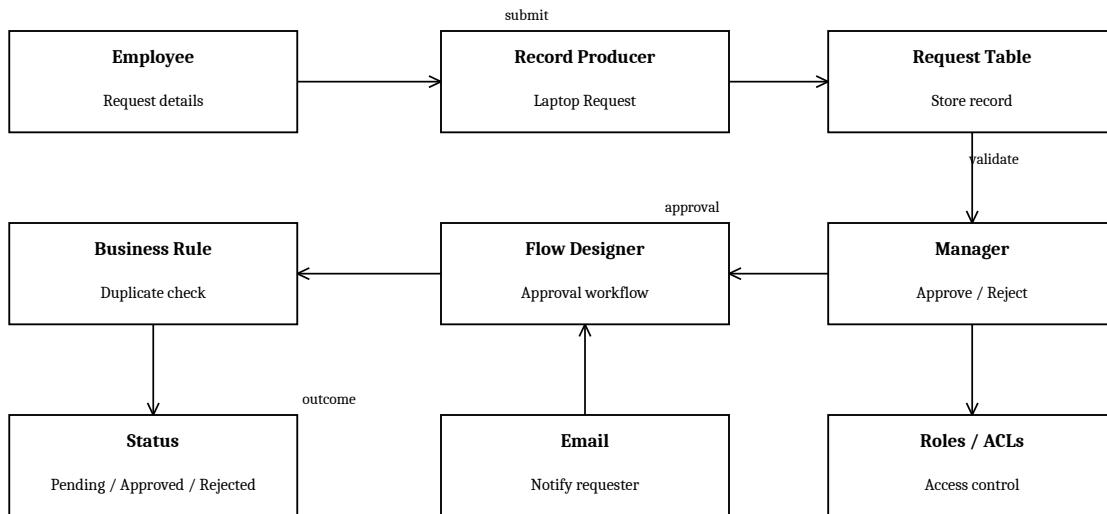
Project Design Phase-II

Data Flow Diagram & User Stories

Data Flow Diagrams:

The DFD shows how a laptop request enters ServiceNow, is stored and validated, moves through manager approval, and returns the outcome through status updates and email notification.

Data Flow Diagram - Employee Laptop Request



User Stories

Use the below template to list all the user stories for the product.

Date	12 September 2026
Team ID	_____
Project Name	Employee Laptop Request Management System
Maximum Marks	4 Marks

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee	Laptop Request Submission	US-1	As an employee, I can submit a laptop request through the Record Producer.	A laptop request record is created with submitted details.	High	Sprint-1
Employee	Request Information	US-2	As an employee, I can provide requester, email, manager, type and reason information.	Required information is captured correctly.	High	Sprint-1

Project Design Phase-II

Data Flow Diagram & User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee	Duplicate Prevention	US-3	As an employee, I can submit only one laptop request in a calendar year.	A second request in the same year is blocked.	High	Sprint-1
Employee	Request Classification	US-4	As an employee, I can select Basic or Advance and provide the request reason.	Selected type and reason are saved correctly.	High	Sprint-1
IT Manager	Manager Approval	US-5	As an IT manager, I can receive and decide on a laptop request approval.	Approval record is created for the manager.	High	Sprint-1
IT Manager	Approval Outcome	US-6	As an IT manager, I can approve or reject a request.	Flow follows the selected approval outcome.	High	Sprint-1
Employee	Status Tracking	US-7	As an employee, I can see Pending Approval, Approved or Rejected status.	Status reflects workflow outcome.	High	Sprint-1
Employee / Manager	Email Notification	US-8	As a stakeholder, I receive email communication for approval and outcome.	Notification is sent to the intended recipient.	High	Sprint-2
IT Employee	Role-Based Access	US-9	As an IT employee, I can create, read and write laptop request records.	ACLs allow permitted operations for it_employee.	High	Sprint-2
IT Manager	Role-Based Access	US-10	As an IT manager, I can view laptop request records without create/modify access.	Read access is allowed and create/write is blocked.	High	Sprint-2
Project Team	Validation & Deployment	US-11	As a project team member, I can validate configurations and update-set capture before deployment.	Business rules, ACLs, flow and application records are validated.	High	Sprint-2

Project Design Phase-II

Requirement Analysis - Customer Journey Map

Date	12 September 2026
Team ID	_____
Project Name	Employee Laptop Request Management System
Maximum Marks	4 Marks
Primary Customer	Employee
Supporting Customer	IT Manager
Scenario	Submitting, validating, approving, tracking and communicating a laptop request
Project	Employee Laptop Request Management System
Journey Type	ServiceNow Laptop Request lifecycle

Project Design Phase-II

Customer Journey Map - Employee Laptop Request

CUSTOMER JOURNEY	ENTICE Discover / understand the request process	ENTER Submit the laptop request	ENGAGE Validate and obtain approval	EXIT Receive outcome and track status	REUSE Maintain traceability for future requests
STEPS	Identify the need for a laptop and understand the request process.	Open the Record Producer and enter requester, email, manager, type, reason and description.	System checks duplicate request policy; Flow Designer starts manager approval.	Manager approves or rejects; status becomes Approved or Rejected.	Request history and controlled records support accountability.
INTERACTIONS	Employee communicates the laptop requirement.	Employee interacts with the ServiceNow Record Producer.	Business Rule validates annual policy; manager receives approval request.	Flow Designer updates the record and sends outcome email.	IT and management use stored records for tracking.
TOUCHPOINTS	ServiceNow catalog / portal entry point.	Laptop Request Record Producer and request fields.	Business Rule; Flow Designer; sysapproval_approver.	Approval status; email notification; request record.	Laptop Request Form table; update set and history.
GOALS & MOTIVATIONS	Help me understand how to request a laptop correctly.	Help me submit complete information without manual email requests.	Help the correct manager review the request and prevent duplicate annual requests.	Help me know whether the request was approved or rejected.	Help the organization maintain traceable and controlled records.
EMOTIONS	Uncertain – ready	Focused – confident	Concerned – supported	Relieved – informed	Confident – accomplished
PAIN POINTS	Manual or verbal request process.	Incomplete details or duplicate requests.	Approval can require follow-up if not automated.	Unclear status or delayed communication.	Poor traceability if requests are handled outside the system.
OPPORTUNITIES	Standardize the laptop request entry point.	Capture consistent requester and request information.	Automate duplicate validation and manager approval.	Automate status updates and email notifications.	Use ServiceNow records, roles and update sets for control.

Project Design Phase-II

Customer Journey Map - Employee Laptop Request

Journey insight:

The journey should minimize manual email or verbal requests, prevent duplicate laptop requests within a year, and preserve clear accountability from submission through manager approval, status updates and email communication. This directly supports the project's goal of a centralized, controlled and traceable Employee Laptop Request Management lifecycle.